

Charlie Wang

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Education

University of Pennsylvania, School of Engineering & Applied Science

Philadelphia, PA

B.S.E. in Digital Media Design

May 2028

- Relevant coursework: Advanced Rendering, Interactive Computer Graphics, Introduction to Algorithms, Discrete Mathematics, Automata and Computability, Computer Animation, Computer Systems
- GPA: 3.86/4.00

Relevant Experience

Tencent Games

Shanghai, China

Game Design Intern (DevOps Department)

May 2025- July 2025

- Led development of an experimental WeChat gardening Mini-Game under senior developer mentorship.
- Independently built a gameplay prototype, demonstrating project feasibility and securing project greenlight.
- Integrated WeChat APIs for gameplay inputs and account-based save data and persistence.
- Designed and implemented core planting, building, and camera systems in unity. Trained coworkers in Unity.

UPGRADE (UPenn Game Development Club)

2024-2026

President, Project Manager

- Ran weekly meetings, hosted workshops, and organized socials and parties.
- Managed the development of UPGRADEKART, a split screen multiplayer party racer with 9 unique karts with special abilities, releasing on Steam in December.
- Oversaw team of 20+ members for UPGRADEKART regarding gameplay programming, game design, and art.
- Spearheaded creation of Game Dev Club Club (an organization of 100+ Game Dev Clubs across the US). GDCC hosted a game jam with 470 participants and 150 submissions over the summer.

Technical Projects

Project Husk | Unity, Shadergraph, HLSL, VFX Graph

2025-Present

- Developed a turn-based tactics game with a strategic multi-action turns and reaction system.
- Designed a modular ability + action architecture with 60+ unique abilities.
- Built a custom lightweight animation system. Authored stylized URP Shaders and VFX for ability effects.
- Built a predictive combat interface that previews movement, action outcomes, resource costs, and incoming enemy actions, with reversible action declarations and action specific inputs.

Mini Minecraft | C++, OpenGL

2025

- Voxel game engine developed in a team of three. Contributed in realistic terrain generation, multithreaded terrain generation, deferred rendering pipeline, and screen space reflections.

Detention Bums | Unity, C#, 2D Art

2026

- Created in a team of two over 7 days for Juniper Dev Game Jam 2026. Placed 25th out of 3500+ entries.
- Lighthearted game combining pen spinning and unit-merging tower defense.

Game Design Projects

Tandus D&D Module & Campaign

2024-2026

- Designed an expanded power progression system for Dungeons & Dragons 5e. Currently publishing.
- 120+ new character abilities built on D&D 5e. 100+ gameplay sessions using & testing Tandus Module.

Game Design Research

2023

- Independent research on racial stereotypes in 20th-century board games; published in Journal of Student Research (2023).

Technical Skills & Interests

Tech: C#, Java, C++, Python, OCaml, Python, HTML/CSS

Software & Tools: Unity, Blender, Maya, Unreal Engine 5, Adobe CC (PS/PP/AE/ID), Git

Interests: Playing games, making games, Dungeons & Dragons and other tabletop games, lifting, manga.